SULIT

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA
MID-TERM EXAMINATION SEMESTER I 2020/2021

SUBJECT CODE	**: SCSR2043/SECR2043/SCSS4213**
SUBJECT NAME	**: OPERATING SYSTEM (PART 1)**
SECTION	**: 01/02/03**
TIME	**: 40 MUNITES**
DATE/DAY	**: 21 DECEMBER 2020**
VENUES	**: 8.00 PM** – 8.40 **PM**

INSTRUCTIONS:

There are TWO parts. Part 1 consists of 30 multiple choice questions and Part 2 consists of 21 multiple choice questions related to computation and critical thinking. This test will contribute 25% towards the total marks of 100%.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section (Circle)	
Lecturer Name (Circle)	

This question paper consists of 9 pages excluding this page.

PART 1

1. Which of the following statements are **TRUE**?

   I. Shortest Remaining Time scheduling may cause starvation.

   II. Pre-emptive scheduling may cause starvation.

   III. Round Robin is better than FCFS in terms of response time.

   A. I, II and III

   B. II and III

   C. I only

   D. I and II

2. How can we disallow circular wait to prevent deadlock from happening?

   A. Make sure non-shareable resources can be accessed simultaneously by various processes.

   B. Make processes acquire resources in priority order.

   C. A process must acquire all the resource it needs before it does anything.

   D. If a process asks for a resource not currently available, block it and take away all of its other resource.

3. The benefits of multithreaded programming can be broken down into four parts **EXCEPT:**

   A. Scalability

   B. Responsive

   C. Dependency

   D. Resource sharing

4. The following statement is related to which condition?

"*Concurrent access to shared data may result in data inconsistency. Maintaining data consistency requires mechanisms to ensure the orderly execution of cooperating processes. This problem occurs in concurrent execution."*

A. Race condition
B. Progress condition
C. Bounded waiting condition
D. Mutual exclusion condition

5. What are the methods to eliminate deadlocks by terminating a process?

I. Abort all deadlocked processes.
II. Preempt some resources from one or more of the deadlocked processes.
III. Terminate one process at a time until the deadlock cycle is eliminated.
IV. Terminate all processes in the unsafe state.

A. I, II, III and IV
B. I and II
C. I, II and III
D. I and III

6. The following are components of a process **EXCEPT**:

A. Scheduling information and program code.
B. Program code and process control information.
C. Context of a program execution and program data.
D. Global variable and context of a program execution.

7. In the producer-consumer problem, the producer needs to synchronize with the consumer for many reasons. Choose the **TRUE** reason(s)?

   I. To allow both producer and consumer to be executed concurrently.

   II. To ensure the consumer will not retrieve data before the producer puts data into the buffer.

   III. To enable buffer to be shared between producer and consumer.

   IV. To solve race condition for two processes.

   A. II, III and IV

   B. I, II and III

   C. I, II and IV

   D. I, III and IV

8. In _____ the processes under consideration must be independent: that is, the order in which they execute must be unconstrained by any synchronization requirements.

   A. deadlock prevention

   B. deadlock detection

   C. deadlock deletion

   D. deadlock avoidance

9. The following statements are true regarding deadlocks **EXCEPT:**

   A. In a multiprogramming environment, several processes may compete for an infinite number of resources.

   B. Deadlock may lead to starvation.

   C. Deadlock is a significant problem that can arise in a community of independent processes.

   D. Deadlock can be avoided by never allowing all four conditions to hold.

10. What is the cycle type for **total**?

```
{
        printf ("Enter 1st number : ");
        scanf ("%d", &number);
        total = b – (b/2) + 5;
        printf ("\n The answer for total = %d", total);
}
```

A. Read burst

B. Write burst

C. I/O burst

D. CPU burst

11. The following figure shows a Peterson solution for two processes P0 and P1 that run concurrently. Suppose P0 just finished executing `flag(0)= TRUE` and P1 executes the `while (flag(0) && turn == 0)`. Which one of the following operations will happen?

Process P0	**Process P1**

`While (TRUE)`
`{`
`  flag[0] = TRUE;`
`  turn = 1;`
`  while (flag[1] && turn == 1)`
`  ; /* do nothing */`
`  critical section`
`  flag[0] = FALSE;`
`  remainder section`
`}`
	`while (TRUE)`
	`{`
	`  flag[1] = TRUE;`
	`  turn = 0;`
	`  while (flag[0] && turn == 0)`
	`  ; /* do nothing */`
	`  critical section`
	`  flag[1] = FALSE;`
	`  remainder section`
	`}`

A. P1 will be busy waiting.

B. P0 will enter its critical section.

C. P1 will enter its critical section.

D. Neither P0 nor P1 enter their critical section.

12. What are the empty and full semaphores indicate in the bounded buffer problem?
    A. Count the number of empty and full buffers.
    B. Count the number of empty and full memory spaces.
    C. Count the number of empty and full queues.
    D. None of the above.

13. A set of instruction that needs to be mutually exclusion with other processes should be put in the _____.
    A. Critical section
    B. Entry section
    C. Exit section
    D. Remainder section

14. In the normal practice, which is the correct sequence in handling deadlock?
    A. Prevention, Allocation, Avoidance, Recovery.
    B. Prevention, Detection, Avoidance, Recovery.
    C. Avoidance, Detection, Prevention, Recovery.
    D. Prevention, Avoidance, Detection, Recovery.

15. Suppose that a process is in “Blocked” state waiting for some I/O service. When the service is completed, it goes to the:
    A. Suspended state
    B. Ready state
    C. Terminated state
    D. Running state

16. The data structures available in the Banker's algorithm are :

A. Available

B. Need

C. Allocation

D. All of the above

17. The following variables that are related to producer consumer problem **EXCEPT**:

A. Critical section

B. Counter

C. Out

D. Buffer

18. If one thread opens a file with read privileges, then _____.

A. Other threads in the another process can also read from that file.

B. Any other thread cannot read from that file.

C. The threads in the same process can also read from that file.

D. All of the above.

19. In semaphore, at the start of the program, semaphore S is always initialized to 1. When semaphore S is 0, what does this mean?

A. All the processes are in remainder section.

B. There is a process in critical section.

C. The process is in remainder section.

D. There is no process in critical section.

20. Which is **FALSE** for CPU scheduler:

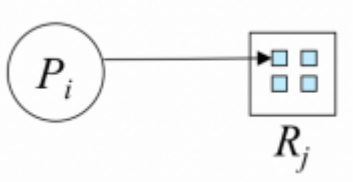
A. Control the degree of multiprogramming.

B. Short-term scheduler is invoked frequently (milliseconds).

C. Known as short-term scheduler.

D. Select which process should be executed next and allocates a CPU.

21. Choose the **TRUE** statement for describing the RAG in the following figure:



A. Process *Pi* requests an instance of resource *Rj*.
B. An instance of resource *Rj* are assigned to *Pi.*
C. Process *Pi* claims an instance of resource *Rj*.
D. All instances of resource *Rj* are assigned to *Pi.*

22. A system is safe if the system can ____ resources to each process in some _____ and still avoid a deadlock.

A. deallocate, order
B. allocate, safe sequence
C. circulate, state
D. preempt, order

23. All of these factors can switch a process except for _____.

A. Memory fault
B. Process scheduler
C. Clock interrupt
D. Supervisor call

24. Which statement is **FALSE,** when this line; *pid = fork()* is executed?

A. Creates a child process where a child process is a copy of the parent.
B. Will return an integer value to variable pid.
C. Parent process has zero value for pid.
D. Parent process has positive value for pid.

25. A semaphore is a shared integer variable:
    A. That cannot drop below zero
    B. That cannot be more than one
    C. That cannot drop below one
    D. That cannot be more than zero

26. Which of the following statements is not valid for the deadlock prevention scheme?
    A. Release all resources before requesting a new resource.
    B. Never request a resource after releasing any resource.
    C. Number the resources uniquely and never request a lower numbered resource than the last one requested.
    D. Request and all required resources be allocated before execution.

27. Termination of the process terminates:
    A. First thread of the process
    B. Last thread of the process
    C. All threads of the process
    D. None of the above

28. The success of CPU scheduling depends on an observed property of processes: process execution consists of a cycle of _____ and _____.
    A. CPU execution, I/O wait
    B. Job scheduler, Process scheduler.
    C. Job scheduler, I/O wait.
    D. CPU execution, CPU scheduler.

29. The time required to create a new thread in an existing process is:

A. Greater than the time required to create a new process.

B. Equal to the time required to create a new process.

C. Less than the time required to create a new process.

D. None of the above

30. The execution of critical sections should satisfy the following requirements:

A. Rap

B. Interrupt

C. Progress

D. Bounded buffer